Part B — Sourcing Strategy & Scale-Up Proposal

DeepThought Business Analytics Internship

Question 2: Building 1000 ICP-Qualified Companies in One Month

The core constraint is quality, not volume. Getting 1000 company names is trivial — getting 1000 genuinely ICP-qualified companies with evidence-backed scores is not. Everything in this plan is designed around that distinction.

The Core Math

Based on the Part A research, roughly 30% of investigated companies pass the Federer filter. That means to get 1000 qualified companies, I need to investigate approximately 3,300–3,500 companies. The plan is built around sourcing that universe efficiently in Week 1 and scoring it systematically in Weeks 2–3.

The Funnel

Stage	Count	Method
Raw universe sourced	3,500	7 sources, Week 1
After hard pre-filters	2,200–2,500	Automated, end of Week 1
After AI scoring (first pass)	1,200–1,400	Claude Haiku, Week 2
After human QA	1,000	Manual review, Week 3–4

Week 1 — Build the Universe (Days 1–7)

Goal: 3,500 unique companies with name, city, segment, and website URL. No scoring yet.

- Days 1–2: Download and parse DSIR list, BSE/NSE SME sector lists, USFDA facility list. These are structured data — fastest to process. Expected yield: ~1,200 companies across target segments.
- Days 2–3: Scrape exhibitor directories from 6–8 target expos (CPHI, BioAsia, Aero India, IMTEX, Agri Intex, ITME). Build Python scraper using BeautifulSoup/Playwright. Expected yield: ~700 companies.
- Days 3–4: Extract industry association directories (IDMA, ACMA, ELCINA, IVAMA, CODISSIA). Query MCA data via Antigravity for target NIC codes and cities. Expected yield: ~800 companies.

- Days 4–5: LinkedIn Sales Navigator extraction — founders/MDs in manufacturing sub-sectors in target cities. Expected yield: ~400 unique companies.
- Days 5–6: Deduplicate master list using fuzzy name matching (Levenshtein distance) and CIN number matching for listed companies.
- Day 7: Website discovery for companies without URLs. Automated Google search: '{company name} {city} site:.com OR site:.in'. Fill gaps manually for high-priority companies.

Week 2 — Automated Scoring (Days 8–16)

Goal: Score all 3,500 companies using AI. Expected output: ~1,200–1,400 first-pass qualifiers.

The scraper: For each company with a website, scrape homepage, /about, /leadership, /products, /news, /careers, and /contact pages. Concatenate to ~8K tokens. Use Playwright to handle JavaScript-rendered sites.

The scoring pipeline:

- Send scraped content to Claude Haiku with a structured ICP scoring prompt
- Receive JSON output: E1/E2 gate results + C3–C8 scores + evidence snippets + confidence flags
- If total score is 40–60 (borderline): re-score with Claude Sonnet for better judgment
- If any criterion has low confidence: flag for human QA in Week 3
- Store all results in a master spreadsheet with full audit trail

Hard pre-filters before AI scoring (saves cost and time):

- No website found after Google search attempt → skip
- Website is a single parked page (total text under 500 characters) → skip
- Company name contains 'Trading', 'Imports', 'Distributors' → skip
- Revenue over Rs.500Cr confirmed from BSE/NSE data → skip

Cost estimate: Haiku first pass ~Rs.0.40 per company × 3,500 = Rs.1,400. Sonnet re-score on ~20% borderline ~Rs.3.50 per company × 700 = Rs.2,450. Total AI cost: approximately Rs.4,000 for the full pipeline.

Week 3 — Human QA (Days 17–24)

Goal: Fix false positives and verify borderline cases. AI scoring has known failure modes that need human correction.

Main AI failure modes to watch for:

- C3 inflation: LLM reads 'ISO 9001' and scores differentiation as moderate. ISO 9001 is a baseline certification, not a differentiator — needs manual correction.

- C4 false positives: Any engineering degree gets scored as technical. A B.Tech from a tier-3 college running a commodity business is not the same as an IIT or ISRO background.
- C6 false positives: LLM reads a press page with 2019 articles and scores growth as moderate. Date-checking needs human eyes.
- C1 false negatives: Company uses 'solutions' and 'services' language but actually manufactures — LLM takes website language too literally.

QA process:

- Auto-QA: Flag companies where C3 evidence mentions only ISO 9001/14001. Flag companies where C6 evidence references news older than 2023. Flag total scores of exactly 40–45.
- Human review: Open website, verify 2–3 key claims, adjust score if needed. Estimated 3–4 minutes per company × 400 flagged = ~25 hours of review over 5 days.
- Spot-check: Review 50 random strong-pass companies to verify AI accuracy.

Week 4 — Final Assembly (Days 25–30)

Goal: Clean final list of 1,000 verified companies plus outreach-ready top tier.

- Days 25–26: Merge clean strong-pass + QA-verified pass. Deduplicate final list. Verify count is at or above 1,000.
- Days 26–28: Add personalization hooks for top 200 companies — the priority outreach tier. These are the A-band companies most likely to convert.
- Days 28–29: Format final deliverable — master CSV with all fields, segment breakdown, city breakdown, score distribution summary.
- Day 30: Buffer for overruns, final QA pass, documentation.

Risk Mitigation

Risk	Mitigation
Yield below 30%	Expand universe: add state industrial estate lists (TSIIC, MIDC, GIDC), Google Maps scraping for industrial areas.
Websites block scraper	Rotate user agents, add delays. Fall back to manual research for high-priority blocked sites.
AI scoring accuracy below 80%	Retune prompt on calibration set of 15 known companies. Split into two steps: Haiku for E1/E2/C5 (factual), Sonnet for C3/C4/C7/C8 (judgment).
QA bottleneck	Tighten auto-QA rules to reduce flagged volume. Recruit second reviewer if needed.

Expected Yield at Each Stage

Week	Output	Yield	Cumulative
Week 1	Raw universe	3,500 companies	3,500
Week 2	AI first pass	~35% pass rate	~1,300 qualifiers
Week 3	Human QA	~75% of flagged pass	~1,100 verified
Week 4	Final list	1,000 confirmed	1,000 + 200 priority

Note on the hand-drawn diagram: The funnel above — Universe → Pre-filter → AI Score → Human QA → Final 1,000 — is what the diagram will show, with tools at each stage and weekly checkpoints marked. Submitted separately via Internshala chat as required.